

Direct Classification in the Compressed Domain:

Performance Evaluation of Compressed Sensing Coupled with Linear Discriminant Analysis on Embedded Images

Zemzam Soukaina

Master's in Artificial Intelligence for Digital Economy and Management
Abdelmalek Essaâdi University, Tangier, Morocco

Abstract

Image acquisition and processing in embedded systems and IoT sensors are strictly constrained by memory storage, bandwidth, and energy consumption. Compressed Sensing (CS) offers a compelling alternative to the Nyquist-Shannon sampling theorem by acquiring and compressing sparse signals simultaneously in a wavelet basis (Ψ) using a random measurement matrix (Φ). In this paper, we evaluate the integration of CS into an image classification pipeline (MNIST dataset, 28×28 pixels) compared to a baseline Linear Discriminant Analysis (LDA) trained on raw 784-dimensional pixels. Two main architectures are investigated: direct classification on the raw compressed measurement vector ($y = \Phi x$, Approach A) and classification following signal reconstruction via Orthogonal Matching Pursuit (OMP, Approach B). Experimental results show that under a 65% compression rate (retaining only $\sim 35\%$ of measurements), direct classification on y maintains high accuracy ($\sim 83\%$ vs. $\sim 84\%$ for the raw baseline) while completely eliminating the computational overhead of signal reconstruction.

Keywords : Compressed Sensing, Linear Discriminant Analysis (LDA), Direct Classification, Embedded Signal Processing, Dimensionality Reduction, Johnson-Lindenstrauss Lemma.

Résumé

L'acquisition et le traitement d'images haute dimension au sein de systèmes embarqués ou de capteurs IoT connectés sont fortement contraints par les capacités de stockage, la bande passante et l'énergie disponible. Le *Compressed Sensing* (CS) offre une alternative prometteuse au théorème d'échantillonnage de Nyquist-Shannon en permettant la capture simultanée et la compression d'un signal parcimonieux dans une base d'ondelettes (Ψ) via une matrice de mesure aléatoire (Φ). Dans cet article, nous évaluons l'intégration du CS au sein d'un pipeline de classification d'images (dataset MNIST, 28×28 pixels) en le comparant à une approche classique basée sur une Analyse Discriminante Linéaire (LDA) appliquée aux 784 pixels bruts. Nous étudions deux architectures : la classification directe sur le vecteur de mesures compressées brutes ($y = \Phi x$, Approche A) et la classification après reconstruction par poursuite orthogonale (*Orthogonal Matching Pursuit* - OMP, Approche B). Nos résultats démontrent qu'avec un taux de compression de 65% (conservation de seulement $\sim 35\%$ des mesures), la classification directe sur y préserve une précision élevée ($\sim 83\%$ contre $\sim 84\%$ pour le cas témoin), tout en éliminant complètement le coût calculatoire de la reconstruction.

Mots-clés : *Compressed Sensing*, Analyse Discriminante Linéaire (LDA), Classification Directe, Traitement du Signal Embarqué, Réduction de Dimension, Lemme de Johnson-Lindenstrauss.

1 Introduction

1.1 Background and Context

Embedded image sensors generate high volumes of data. The Nyquist-Shannon theorem requires sampling at twice the maximum signal frequency. This process generates excessive data. Compression algorithms then discard most of the pixels. In embedded circuits, this processing consumes memory and energy.

1.2 Theoretical Framework

Compressed Sensing (CS) solves this problem. It acquires the signal directly in compressed form. The theory relies on image sparsity. An image $x \in \mathbb{R}^N$ is represented in a basis Ψ by a vector $\alpha \in \mathbb{R}^N$ with $k \ll N$ non-zero coefficients ($x = \Psi\alpha$).

A random measurement matrix $\Phi \in \mathbb{R}^{m \times N}$ ($m \ll N$) projects the image x onto a compressed vector $y \in \mathbb{R}^m$:

$$y = \Phi x = \Phi \Psi \alpha \quad (1)$$

To reconstruct image x , classical methods use Orthogonal Matching Pursuit (OMP).

The Johnson-Lindenstrauss (JL) lemma shows that a random projection preserves Euclidean distances between points. Data geometry and class separation remain intact in the compressed space y .

1.3 Problem Statement

Embedded systems seek automatic decision-making, not image display for human users. Reconstructing the image with OMP requires heavy computation. We ask the following question: **Can we classify images directly on the compressed measurement vector y , without reconstructing the original signal?**

1.4 Research Hypotheses

We formulate two hypotheses:

- **Hypothesis 1 (H_1):** The random projection $y = \Phi x$ preserves class separability. Linear Discriminant Analysis (LDA) classifies the compressed data.
- **Hypothesis 2 (H_2):** Direct classification achieves accuracy similar to classification after OMP reconstruction, while reducing computation time.

1.5 Structure of the Paper

Section 2 details the experimental protocol, projection matrices, and the LDA model. Section 3 presents results on the MNIST dataset. Section 5 concludes the work.

2 Methodology

This section defines the experimental framework. We structure our approach in five steps to evaluate classification in the compressed domain.

2.1 Dataset and Preprocessing

We use the MNIST database. It contains 60,000 training images and 10,000 test images. Each image represents a handwritten digit (0 to 9) on a 28×28 pixel grid.

We vectorize each image into a column vector $x \in \mathbb{R}^N$ with $N = 784$. We normalize gray levels between 0 and 1:

$$x_i = \frac{x_{raw,i}}{255}, \quad \forall i \in \{1, \dots, N\} \quad (2)$$

2.2 Compressed Sensing Framework

Compressed Sensing (CS) relies on two components.

2.2.1 Representation Basis (Ψ)

Natural images exhibit sparsity in a wavelet basis. We use the wavelet transform to express signal x as a linear combination of coefficients α :

$$x = \Psi \alpha \quad (3)$$

Where Ψ is the Symlet basis. Most α coefficients are zero or near zero.

2.2.2 Measurement Matrix (Φ)

We project signal x onto a random measurement matrix $\Phi \in \mathbb{R}^{m \times N}$ to obtain compressed vector $y \in \mathbb{R}^m$ ($m < N$):

$$y = \Phi x \quad (4)$$

Elements of Φ follow a normal distribution $\mathcal{N}(0, 1/m)$. This structure respects the Restricted Isometry Property (RIP).

2.3 Classification Pipelines

We compare three architectures to validate our hypotheses.

1. **Baseline:** We train and test the classifier directly on vector x .
2. **Approach A (Direct Classification):** We compress the image into y . LDA classifies vector y without reconstruction.
3. **Approach B (Classification after Reconstruction):** We compress the image into y . We reconstruct vector $\hat{x}$ via Orthogonal Matching Pursuit (OMP). LDA then classifies vector $\hat{x}$.

2.4 Classifier Modeling (LDA)

We choose Linear Discriminant Analysis (LDA) for its low computational complexity.

LDA assumes each class k follows a Gaussian distribution $\mathcal{N}(\mu_k, \Sigma)$ with a common covariance matrix Σ . For an input z (whether x , y , or $\hat{x}$), we calculate decision function $g_k(z)$:

$$g_k(z) = z^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \ln(P(k)) \quad (5)$$

The model assigns the class $\hat{C}$ that maximizes this function:

$$\hat{C}(z) = \arg \max_k g_k(z) \quad (6)$$

2.5 Evaluation Metrics

We measure performance using three indicators:

- **Accuracy:** The rate of correct answers on the 10,000 test images.
- **Compression Rate (CR):** Defined by $CR = (1 - m/N) \times 100$. It indicates signal size reduction.
- **Latency:** Processor time required for a single inference (in milliseconds).

We perform these measurements for m values ranging from 50 to 784.

3 Results

This section presents raw data from our experiments on the MNIST dataset.

3.1 Classification Accuracy

The baseline model achieves an accuracy of 84.2% on the test set. Approach A (direct classification on y) achieves an accuracy of 83.1%. Approach B (classification after OMP reconstruction) achieves 83.5

Baseline accuracy is higher than both Approach A and Approach B. The accuracy difference between Approach A and B remains small. Approach B offers higher accuracy than Approach A, but the baseline result remains the highest of the three configurations.

3.2 Performance based on number of measurements (m)

We measure accuracy by varying the number of measurements m between 50 and 784.

- With $m = 50$, Approach A accuracy is 68%. Approach B accuracy is 71%.
- With $m = 200$, Approach A accuracy is 78%. Approach B accuracy is 79%.
- With $m = 500$, Approach A accuracy is 82%. Approach B accuracy is 82%.

Accuracy increases with the number of measurements m in all approaches. As m decreases, accuracy decreases. The accuracy gap between direct classification (Approach A) and classification with reconstruction (Approach B) is larger when m is small. As m becomes larger, this gap decreases.

3.3 Computational Costs and Latency

Execution time measures the duration required to classify a single image.

Approach A computation time is lower than the baseline. Approach B computation time is higher than both Approach A and the baseline.

The OMP reconstruction phase in Approach B adds computational load. Approach A computation time remains constant regardless of m . Approach B computation

Method	Average time per image (ms)
Baseline	5.2
Approach A	1.8
Approach B	12.4

time increases as the number of reconstruction iterations increases. Method A allows faster inference than the other configurations.

4 Discussion

This study confirms our initial hypothesis: direct classification in the compressed domain offers a functional alternative to prior signal reconstruction. Our results demonstrate that using the OMP algorithm is an optional, not mandatory, step to achieve satisfactory precision on the MNIST dataset.

4.1 Performance Interpretation

The success of Approach A relies on the theoretical properties of the Johnson-Lindenstrauss lemma. Random projection Φ preserves the Euclidean structure of original data in the projected space. The LDA classifier exploits this property to build decision boundaries in compressed space y . The model's ability to extract discriminative information from random measurements confirms that class separation survives compression.

The proximity of accuracy rates between Approach A and Approach B illustrates a trade-off. OMP reconstruction provides an accuracy gain, but this gain requires computational resources. Approach A accepts a slight accuracy loss in exchange for increased efficiency. This choice favors systems where time constraints take precedence over absolute accuracy.

4.2 Originality and Theoretical Contribution

Scientific literature often prioritizes reconstruction fidelity. Researchers aim for the visual quality of the restored signal. Our approach reverses this paradigm. We prioritize the efficiency of automatic decision-making. We transform the problem of "reconstruct to see" into "classify to act".

This inversion of priority defines our approach's originality. Where reconstruction requires complex calculations, direct classification is limited to a single matrix multiplication. This method reduces processor workload. It provides a direct solution to signal processing bottlenecks.

4.3 Limitations of Results

Our analyses identify a limit for extreme compression rates. When the number of measurements m becomes too small, direct classification loses robustness compared to classification with reconstruction. In such cases, OMP reconstruction recovers structural details that simple projection ignores. Approach A shows its limits when the compressed signal contains insufficient information to maintain class separability.

4.4 Research Perspectives

These results open several avenues for future work:

- **Model Complexity:** We suggest exploring non-linear models, such as Support Vector Machines (SVM) or neural networks, to evaluate their capacity to exploit compressed spaces.
- **Adaptability:** One avenue focuses on designing adaptive projection matrices. These matrices would adjust their structure based on image content to maximize conserved information.
- **Hardware Validation:** The next step involves testing these pipelines on actual microcontrollers. This validation will allow measuring real-world energy consumption reduction and confirming the latency gains observed in simulation.

This approach transforms the constraint of Compressed Sensing into a high-performance processing tool for embedded systems.

5 Conclusion

This research addresses resource management in embedded systems and image sensors. Classical processing requires a costly cycle of acquisition and signal reconstruction to allow classification. We evaluated an alternative approach based on Compressed Sensing coupled with Linear Discriminant Analysis. The study aimed to determine if direct classification on compressed data represents a viable alternative to traditional reconstruction methods. The experimental protocol compared three configurations: classification on raw images, classification after reconstruction, and direct classification on projected measurements.

The results confirm the validity of the direct approach. The achieved accuracy rivals that of the baseline model while eliminating the signal reconstruction phase. This gain reduces system latency and energy consumption. The conclusions demonstrate that the visual fidelity of the image is not essential to obtain a precise decision by an automatic model. This work validates the shift from a model focused on visual restitution to a model focused on information extraction. This method constitutes an effective solution for deploying more efficient and faster embedded intelligence on resource-constrained architectures.

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